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Dynamic Co(II)/Co(III) Cycle Driven by Outer- and Inner-Sphere Electron Transfer for Sustained Peroxymonosulfate Activation

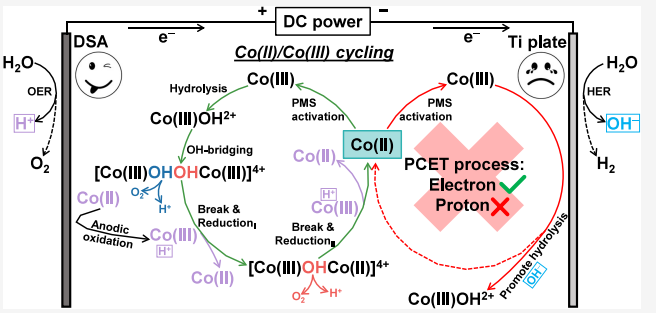
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ABSTRACT: In the realm of transition metal (M^{n+}) activated peroxymonosulfate (PMS), sluggish reduction kinetics of $M^{(n+1)+}$ often lead to the rapid deactivation of catalytic centers, posing a significant challenge for commercialization of homogeneous advanced oxidation processes (AOPs). We report a pioneering elucidation of a distinct Co(II)/Co(III) cycling mechanism within electrochemically enhanced PMS-AOPs, utilizing Co(II) as a model catalyst. Remarkably, this cycling process predominantly unfolds in the anodic region, rather than the cathodic, revealing a novel aspect of electrochemical modulation. Co(III), generated by anodic oxidation, emerges as a pivotal species that disrupts the dimerized hydrolysis product ($[Co(III)OH]_2^{4+}$). Electron transfer from the hydroxyl oxygen in $[Co(III)OH]_2^{4+}$ to Co(III) induces electron redistribution, ultimately facilitating Co(III) reduction and release via both outer- and inner-sphere electron transfer pathways. Gibbs free energy calculations unequivocally confirm the spontaneity of the cyclic process. Our system exhibits superior performance metrics, achieving Co(IV)═O (5.57×10^{-2} mM/M Co) and $SO_4^{\bullet-}$ (2.51×10^{-6} mM/M OSO_3) yields that surpass most reported catalytic systems, along with an exceptional mass activity of Co(II) (368.87 L/g). This study offers a fresh perspective on M^{n+} regeneration for sustained PMS activation in homogeneous transition metal catalysis, with potential implications for advancing the field of environmental remediation.



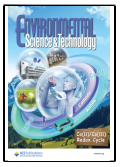
KEYWORDS: *electrochemical AOPs, PMS activation, transition metal, catalytic regeneration, Co(II)/Co(III) cycle*

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INTRODUCTION

As an indispensable basic resource and crucial environmental medium, water resources have gained paramount importance for human survival. Antibiotics, emblematic of emerging pollutants, are widely detected in aquatic environments, giving rise to severe ecological risks and undermining the utility of water resources. Thus, there is an urgent imperative to develop economically viable and effective purification technologies. Among the various approaches, peroxymonosulfate-based advanced oxidation processes (PMS-AOPs) have garnered considerable attention in the field of water treatment due to their ability to completely destroy harmful pollutants. The efficacy of Fenton-like reactions hinges on the interaction between PMS and active catalytic centers, facilitating electron transfer and the subsequent generation of reactive species.[1−6] Transition metals, Co, Fe, Cu, and Mn,[7−10] stand out as exceptional activators for PMS due to their high catalytic efficiency and ease of manipulation. Particularly, Co(II) exhibits remarkable activity in activating PMS even at trace levels (parts per billion), positioning it as a highly promising candidate for initiating Fenton-like reactions. The widely accepted activation mechanism is delineated in eqs S1−S4 (Text S1).[11] Co(II) regeneration occurs through a single-electron transfer reaction between Co(III) and HSO_5^-. However, the reaction rate between Co(III) (0.35 M^{-1} s^{-1}) and PMS is substantially slower compared to that of Co(II) (100 M^{-1} s^{-1}),[12] especially in trace Co(II) concentrations.[13] This process also accompanied by the generation of $SO_5^{\bullet-}$ with a relatively low redox potential (1.1 V).[14] Consequently, eq S4 has been identified as the rate-limiting step in the Co(II)/Co(III) cycle, necessitating higher doses of Co(II) to achieve the desired catalytic oxidation effect. This is unacceptable, given the potential for one-time use of Co(II) in PMS-AOPs and the ensuing risk of secondary environmental pollution from excessive Co(II) levels. To maximize catalytic activity while efficiently utilizing limited resources, achieving rapid catalyst cycling is imperative. Therefore, overcoming the kinetic barrier of Co(III) reduction to accelerate the Co(II)/Co(III) cycle is imperative for achieving sustainable catalysis,

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marking a significant advancement in water purification technologies.

Extensive endeavors have been devoted to identifying electron donors that can alleviate the challenge posed by Co(III) reduction. Current studies enhance Co cycling in Co(II)/PMS systems by introducing electron donors such as small organic acids[15] and phosphate,[16] which form complexes with Co(II) to optimize its electronic structure. Various cocatalysts such as bicarbonate,[17] oxalate,[18] molybdenum,[19] hydroxylamine,[20] polyphenols,[9] and ethylenediamine-*N*,*N*′-disuccinic acid[21] have been explored to improve the redox cycling of transition metal ions and bolster their catalytic activity. The majority of these electron donors are employed at concentrations two orders of magnitude higher than that of trace Co(II), albeit they can elevate the Co(III) reduction rate to 0.35−25 M^{-1} s^{-1}.[22] However, the introduction of any ligand appears to generate toxic byproducts or potential secondary pollution. Therefore, it is imperative to develop efficient and environmentally benign methods for electron supply to achieve the effective conversion between different valence states of Co, thereby enhancing the sustained activation and high utilization of PMS.

It is widely recognized that electrochemical processes present a promising avenue for addressing this challenge. Specifically, the driving force can be precisely controlled by adjusting the applied potential, enabling controllable acceleration of the kinetics process. Concurrently, enhanced mass transfer at the electrode surface overcomes diffusion limitations. More importantly, compared to traditional chemical reductants, electrochemistry directly utilizes sustainably supplied electrons as a clean reductant, minimizing unnecessary byproducts and side reactions. Additionally, the *in situ* generation of reactive oxygen species (ROS), direct electron transfer, and PMS cathodic activation can all be utilized for the oxidative degradation of organic pollutants.[23−25] This constitutes the distinctive advantage of electrochemical strategies that sets them apart from other AOPs. However, reports on the application of electrochemistry for transition metal reduction in PMS-AOPs, particularly those involving homogeneous catalysis, have been poorly reported and the underlying mechanisms remain elusive. Notably, the environment conditions necessary for Co(III) reduction have largely been overlooked. It is crucial to emphasize that Co(III) is highly prone to hydrolysis (with a first hydrolysis constant K_{h1} = 0.22 ± 0.05), resulting in minimal free Co(III) at pH levels exceeding 2.0.[26] Hydrolyzed Co(III) becomes encapsulated by OH^- ions, rendering it difficult to reduce at the cathode. Furthermore, evidence suggests that the direct reduction of Co(III) necessitates the involvement of protons.[16] However, this process is hindered at the cathode due to the hydrogen evolution reaction, which enriches the cathode's vicinity with OH^- ions.[27] On the contrary, the oxygen evolution reaction occurring at the anode produces an acidic H^+-rich interfacial microenvironment, which seems to favor the presence and reduction of Co(III). Consequently, the authentic pathway of Co cycling during electrochemical processes remains obscure. A comprehensive investigation into the $M^{n+}/M^{(n+1)+}$ (where M denotes the metal species) cycling mechanism within electrochemical processes holds dual promise: it could significantly enhance the catalytic activity of M^{n+} in PMS-AOPs and pave the way for efficient integration of electrochemistry with PMS-AOPs in water treatment applications. Such insights would undoubtedly contribute to advancing the field and fostering innovative solutions for environmental remediation.

Basically, the primary objective of this study is to elucidate the underlying mechanism of Co cycling within the Co(II)/PMS system during electrochemical processes, with a particular focus on optimizing the catalytic activity of transition metals. To this end, oxidative degradation efficiency of the electrochemical/cobalt(II)/peroxymonosulfate (EC/Co(II)/PMS) system was comprehensively assessed across a range of conditions and environmental parameters, as well as against various common organics. A rigorously investigation into the generation and transformation of ROS was conducted utilizing scavenging experiments, chemical probes, and electron paramagnetic resonance (EPR) spectroscopy to confirm Co(II) as the active source. Possible Co cycling in the system was elucidated through multiple cycle tests and EDTA trapping. Furthermore, the pivotal reaction pathways implicated in the Co cycling process were meticulously explored through a combination of diverse electrochemical analytical techniques, oxygen concentration determination, *in situ* confocal Raman spectroscopy, and density functional theory (DFT) calculations. This multifaceted approach led to the proposal of a novel cycling mechanism. Ultimately, this research offers deeper insights into the redox behavior of metal ions in PMS-AOPs that involve homogeneous transition metal catalysis, contributing to a more nuanced understanding of this important field.

MATERIALS AND METHODS

Materials and Chemicals, Experimental Procedures, and DFT Calculation. Detailed information for chemicals used in this work is listed in Text S2 of the Supporting Information. All reagents were at least of analytical grade and were used without further purification. Detailed information on experimental procedures and DFT calculation are provided in Texts S3 and S4, respectively.

Analytical Methods. The concentration of organic compounds was analyzed using high-performance liquid chromatography (HPLC). ROS were identified using an electron paramagnetic resonance (EPR) spectrometer. The PMS concentration was measured using an ultraviolet−visible (UV−vis) spectrophotometer. Electrochemical tests were conducted using an electrochemical workstation. The dissolved oxygen concentration in the reaction solution was measured with a dissolved oxygen meter. *In situ* Raman spectra were collected using a confocal Raman spectrometer. Detailed information is provided in Text S5.

RESULTS AND DISCUSSION

Catalytic Performance Assessment and ROS Generation Mechanism. *Catalytic Performance of the EC/Co(II)/PMS System.* The catalytic degradation performance of the EC/Co(II)/PMS system was assessed using ciprofloxacin (CIP), a widely utilized broad-spectrum antibacterial agent, as a model pollutant (see physicochemical properties in Table S1). The EC/Co(II)/PMS oxidation process achieved over 95% degradation of CIP within 30 min, compared to approximately 53% of CIP by the Co(II)/PMS system alone (Figure S1a). This finding indicates that electrochemistry acts as a cocatalyst, enhancing the catalytic activity of Co(II)/PMS. The degradation of CIP followed pseudo-first-order kinetics (Figure S1c), with the kinetic constant (k_{obs}) for EC/Co(II)/

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Environ. Sci. Technol. 2025, 59, 19570−19581

PMS (0.081 min^{-1}) being 3.21 times that of Co(II)/PMS (0.025 min^{-1}) (Figure S1d). In addition, direct electron transfer (DET), potential generation of high-valent cobalt-oxo (Co(IV)=O) by the anode, and cathode-activated PMS contributed minimally to CIP removal (Figure S1b). The primary consumption of PMS was attributed to activation by Co(II) and the cathode, as well as CIP oxidation (Figure S2a). Notably, the EC/Co(II)/PMS system achieved a PMS utilization rate of 92.4%, maximizing its effectiveness (Figure S2b). CIP itself had minimal impact on PMS decomposition, and the PMS consumption rate in the EC/Co(II)/PMS process (0.082 min^{-1}) significantly exceeded those of EC/PMS (0.012 min^{-1}), Co(II)/PMS (0.028 min^{-1}), and their combined rate (Figure S2c). This indicates that electrochemistry effectively promotes cobalt cycling, enhancing PMS decomposition. Optimal Co(II) and PMS concentrations were determined to be 6 and 90 μM, respectively, in batch experiments (Figures S3−S7). Importantly, the Co(II) concentration used was only approximately 1/3 of the national discharge standards (GB 25467-2010) and the U.S. Environmental Protection Agency's secondary drinking water standards (1 mg/L, 17 μM). Under these optimal conditions, the energy consumption of the EC/Co(II)/PMS process was 1.76 kWh m^{-3}, representing at most 15.13% of the control group's consumption (Table S2). The initial pH had the most significant impact on the catalytic activity of the EC/Co(II)/PMS, primarily due to H^+ concentration and Co species present (detailed analysis in Figure S5). However, the EC/Co(II)/PMS system maintained strong catalytic degradation capability across a wide pH range (4−10). Furthermore, compared to PDS, H_2O_2, and PAA, EC/Co(II) was more effective at activating PMS to generate reactive oxygen species (ROS) for CIP degradation (detailed analysis in Figure S8).

Generation and Transformation Pathway of ROS. To further semiquantify the roles of ROS in CIP degradation, scavenging experiments were conducted[28] (detailed analysis in Figures S9 and S10). The results indicated that multiple ROS were present in the EC/Co(II)/PMS system, including $O_2^{\bullet-}$, $^1O_2$, •OH, $SO_4^{\bullet-}$, and Co(IV)=O, and were rigorously validated through EPR spectroscopy and chemical probes (detailed analysis in Figures S11−S15). Combined with the EPR spectrum of the EC/PMS system (Figure S11b), it was shown that •OH was generated via water oxidation at the anode and PMS activation at the cathode.[29,30] Also, •OH will be generated during PMS activation by Co(II). $SO_4^{\bullet-}$ is unequivocally produced through the activation of PMS by Co(II) and the cathode. Given the known interconversions between H_2O_2, •OH, $^1O_2$, and $O_2^{\bullet-}$,[31] the scavenging experiments alone cannot definitively quantify the oxidative contributions of each ROS. Based on previous research, we summarized the transformations into 10 pathways (pathways I−X)[31−35] (Figure S16 and Table S3). Notably, no H_2O_2 was generated during the EC/Co(II)/PMS process (Figure S17), implying that the generation of $^1O_2$ involves pathways VI, IX, and X, while the generation of $O_2^{\bullet-}$ involves pathways II and III. Although $^1O_2$ can form via the self-decay of PMS (pathway X, eq S57), the efficiency of producing $^1O_2$ via self-decomposition of uncatalyzed PMS is relatively low, with a reaction rate constant k_1 of 0.2 M^{-1} s^{-1},[24,36] as supported by the EPR spectrum (Figure S15c). To substantiate the production of $^1O_2$ via the reaction of $O_2^{\bullet-}$ and •OH, FFA was employed as a chemical probe with a high reaction constant for $^1O_2$ (1.2×10^7 M^{-1} s^{-1}).[37] The addition of *p*-BQ or TBA inhibited FFA degradation in the EC/Co(II)/PMS system (Figure S18), and a robust negative linear relationship was observed between *p*-BQ/TBA concentration and the rate constant of FFA degradation on a natural logarithmic scale (Figures S18). Furthermore, the absence of a TEMP−$^1O_2$ characteristic peak in the EPR spectrum in the presence of *p*-BQ/TBA provided additional evidence (Figure S19). The generation of $^1O_2$ in the system was further demonstrated by 9,10-anthracenediyl-bis(methylene)dimalonic acid (ABDA) as an indicator[38,39] (Figure S20). $O_2^{\bullet-}$ is likely generated through PMS/O_2 reduction at the cathode.[29] In fact, introducing O_2 near the cathode did not enhance FFA degradation, and N_2 did not inhibit the production of $^1O_2$ (Figure S21), suggesting that PMS reduction at the cathode produces $O_2^{\bullet-}$. This was confirmed by the detection of a DMPO-$O_2^{\bullet-}$ signal peak in the EPR spectrum of the EC/PMS system (Figure S14) and the specific conversion of nitrotetrazolium blue chloride (NBT) to monoformazan (MF) (Figure S22).

Some studies have corroborated that high-valent metal-oxo species constitute the predominant active species in Fenton-like reactions.[40,41] To elucidate the contribution of Co(IV)=O to CIP degradation within our reaction system, PMSO was employed as a specific quencher.[42] The introduction of 90 mM PMSO into the EC/Co(II)/PMS system led to a near-total cessation of CIP degradation, with this effect intensifying as the PMSO concentration rose (Figure S23). Consequently, Co(IV)=O is postulated as the primary reactive species. The identification of methyl phenyl sulfone ($PMSO_2$) during PMSO oxidation in the EC/Co(II)/PMS system provides further substantiation of this hypothesis (Figure S24).[43] This observation suggests the occurrence of an oxygen atom transfer reaction, a mechanism commonly implicated in the detection of high-valent metal-oxo species. In contrast, the EC/Co(II)/PMS system demonstrated substantial PMSO consumption and $PMSO_2$ production, where the near-perfect conversion rate from PMSO to $PMSO_2$ was almost 100% throughout the degradation process (Figure S24b). This compellingly underscores the pivotal role of Co(IV)=O species within the EC/Co(II)/PMS system. In addition, a comparable, significant conversion of PMSO to $PMSO_2$ was observed in the EC/Co(II) system, indicating that the electrochemical process can also facilitate the oxidation of Co(II) via a double electron transfer to yield Co(IV)=O. Co(IV)=O is prone to engaging in oxygen atom exchange reactions with H_2O.[40,44] The detection of $PMS^{16}O^{18}O$ with a mass-to-charge ratio (m/z) of 159.0368 in the ^{18}O isotope labeling experiment provides further definitive evidence for the involvement of Co(IV)=O in the reaction (detailed analysis in Figure S25). Collectively, the results from the PMSO probe experiments and the ^{18}O isotope labeling technique offer robust evidence for the formation of Co(IV)=O in the EC/Co(II)/PMS process.

Oxidation Contribution Calculation of ROS for CIP. To elucidate the oxidation mechanism underlying the EC/Co(II)/PMS process, we employed competitive kinetics to quantify the oxidative contributions of various ROS. The steady-state concentrations of active species, namely $SO_4^{\bullet-}$, •OH, $^1O_2$, and Co(IV)=O, were meticulously determined using BA, FFA, and PMSO as chemical probes, in conjunction with established second-order rate constants (Table S4 and Figures S26−S28). The steady-state concentration of Co(IV)=O attained 3.34×10^{-10} M, exceeding those of $SO_4^{\bullet-}$, •OH, and $^1O_2$ by two to four orders of magnitude (Figure S28a). Subsequently, the contributions of individual oxidation pathways were assessed

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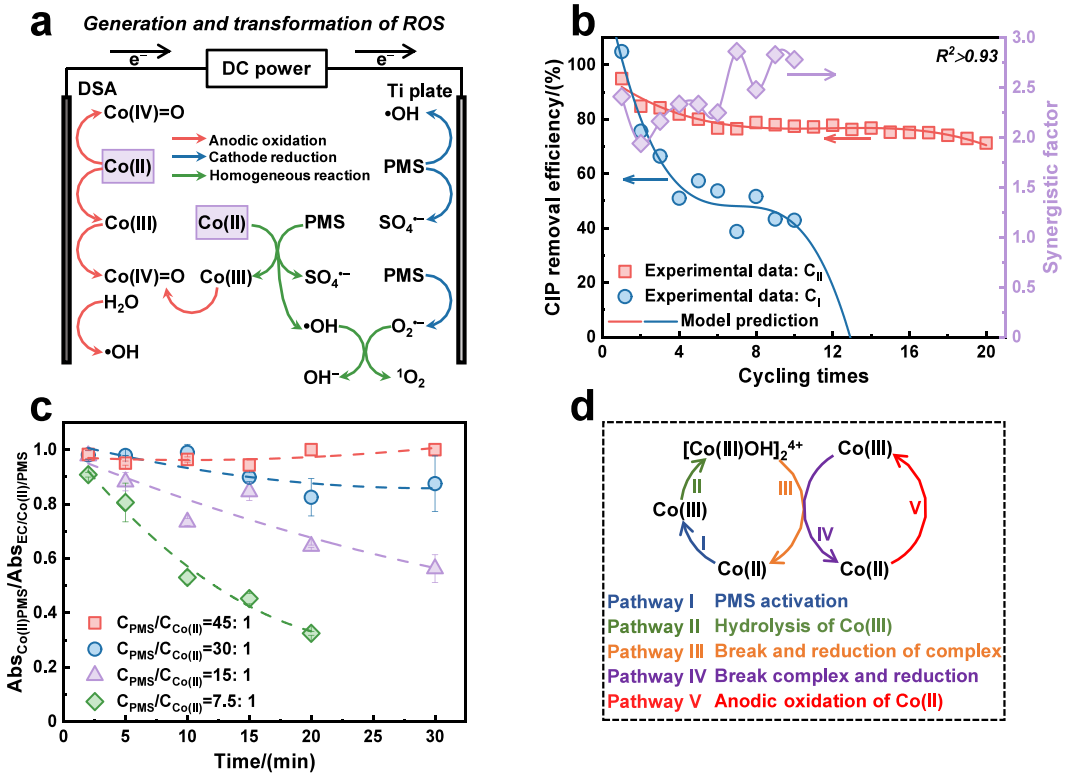


Figure 1. Proposed apparent Co cycling pathway. (a) Potential major ROS generation and transformation pathways in the EC/Co(II)/PMS system. (b) Trend comparison for CIP degradation by sum (C_I) of EC/Co(II), EC/PMS, and Co(II)/PMS systems vs the EC/Co(II)/PMS system (C_{II}) in cycling tests and synergistic factor of the EC/Co(II)/PMS system. (c) Co(III)-EDTA absorbance ratio obtained by EDTA capture of Co(II)/PMS and EC/Co(II)/PMS systems under different $C_{PMS}/C_{Co(II)}$ ratios. (d) Schematic representation of the Co(II)/Co(III) transformations involved in Co(II)/PMS and EC/Co(II)/PMS systems. Reaction conditions: [CIP] = 15 μM, [PMS] = 90 μM, Co(II) = 6 μM, Na_2SO_4 = 50 mM, current density = 10 mA/cm^2, T = 293 K, initial solution pH = 6.00 ± 0.05, if needed, except Co(II) = 0.3 mM for c.

based on their respective steady-state concentrations, revealing a decreasing order of importance as follows: Co(IV)=O > $^1O_2$ > $SO_4^{\bullet-}$ > •OH > direct electron transfer > PMS oxidation (Figure S28b). In summary, the degradation of CIP is attributed to multiple oxidation pathways involving ROS, with the nonradical pathway emerging as the predominant oxidation route. Notably, Co(IV)=O has been identified as the key active species in the EC/Co(II)/PMS process, owing to its high steady-state concentration and significant oxidative contribution.

True Cycling Origin of Co(II)/Co(III). From the aforementioned facts, it can be inferred that Co(II) facilitates the generation of $SO_4^{\bullet-}$, •OH, and Co(IV)=O, with •OH being directly conducive to the production of $^1O_2$. Although electrochemical processes can also generate a small amount of ROS contributing to CIP degradation, it remains undeniable that Co(II) serves as the key active source in the catalytic reaction (Figure 1a). This conclusion has been comprehensively evaluated and demonstrated through degradation kinetics and the investigation of ROS generation and transformation. Therefore, Co(II) emerges as a pivotal active source for catalytic reactions, and the rejuvenation of Co(II) ensures the sustained and efficient progression of the catalytic oxidation reaction. The EC/Co(II)/PMS system achieved average CIP removal rates of 81.34 and 78.26% over 10 and 20 cycles, respectively. Furthermore, the cumulative CIP degradation (C_I) across each cycle was calculated for the EC/Co(II), EC/PMS, and Co(II)/PMS systems, revealing a markedly lower trend compared to the EC/Co(II)/PMS system (C_{II}) (Figure 1b and Figure S29). This disparity may serve as compelling evidence for cobalt cycling. It is noteworthy that C_I encompasses twice the direct electron transfer oxidation of CIP by electrochemical anodes, implying that the benefits derived from cobalt cycling are even more pronounced. The Co(II)/Co(III) cycling process was further validated through UV−visible absorption spectroscopy of Co(III)-EDTA[15] (Figure 1c and Figures S30 and S31).

In Co(II)/PMS systems, Co(II) typically undergoes transformation into Co(III) upon PMS activation. However, it is surprising that the absorbance of Co(III)-EDTA does not increase (Figure S31), indicating that Co(III) does not accumulate but rather decreases steadily during the reaction. This phenomenon can be attributed to the ease of Co(III) hydrolysis, characterized by a first hydrolysis constant (K_{h1}) of 0.22 ± 0.05. The deprotonated Co(III) species, [Co(III)-OH]$^{2+}$, rapidly dimerizes to form a binuclear Co(III) species bridged by two OH groups ([Co(III)OH]$_2^{4+}$).[11,26] This species is difficult to stabilize and capture by EDTA. Indeed, even when K_{h1} is calculated at 0.17, over 99% of Co(III) exists in a dimer form at pH exceeding 2.6 (Figure S32a). This conclusion aligns with the relationship between Co(III) species and pH obtained through software simulations (Figure S32b). The production of the [Co(III)OH]$_2^{4+}$ complex involves the displacement of H_2O coordination, which obstructs the outer-sphere electron transfer process and ultimately leads to a cathodic reduction that may not constitute the primary recycling pathway. This process is mainly governed by the chemical properties of Co(III) and is

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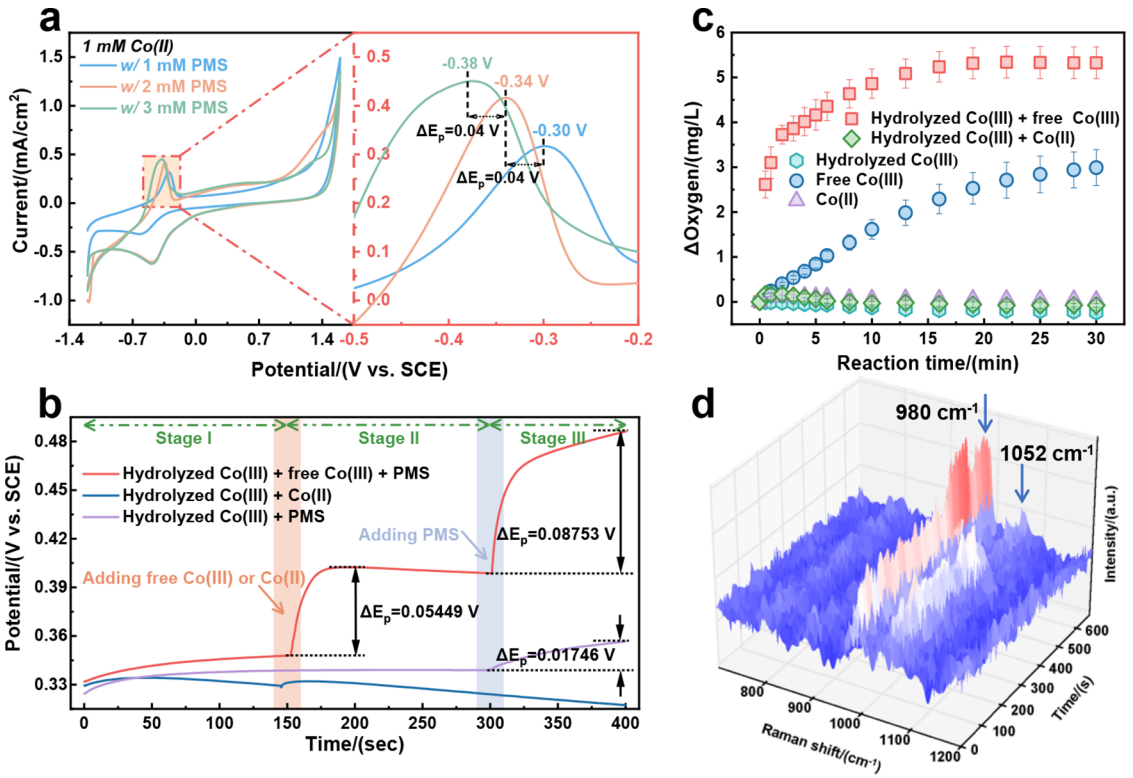


Figure 2. Cyclic mechanism of Co(II)/Co(III). (a) Cyclic voltammetry (CV) curves of the effect for PMS concentration on the redox potential of Co(II). (b) Open-circuit potential (OCPT) curves of reactions between different species. (c) Evolution of oxygen concentration in different reactions. (d) *In situ* confocal Raman spectroscopy of the reaction process between $[Co(III)OH]_2^{4+}$ and free Co(III). Reaction conditions: Co(II) = free Co(III) = [PMS] = 1 mM for a, b, and d, if needed, free Co(III) = 0.3 mM for c.

independent of the electrode material. Detection of Co(III)-EDTA revealed that, in addition to most undergoing hydrolysis, a small fraction of Co(III) generated by Co(II)-activated PMS remains in its free form. The decrease in free Co(III) over time can be attributed to continuous hydrolysis of Co(III) on the one hand, and on the other, the activation of PMS may result in the production of more Co(III), which further promotes the hydrolysis of free Co(III). The Co(II)/Co(III) conversion in the Co(II)/PMS system primarily involves pathways I and II (Figure 1d). As the PMS/Co(II) concentration ratio ($C_{PMS}/C_{Co(II)}$) rises, the Co(III)-EDTA signal of the Co(II)/PMS system intensifies (Figure S31). This is because excess PMS converts more Co(II) into Co(III) within the same time interval. A similar decrease in Co(III)-EDTA absorbance was observed at constant $C_{PMS}/C_{Co(II)}$ during the electrochemical process, but this was not due to electrosorption or precipitation, as the total cobalt concentration in the EC/Co(II)/PMS system remained nearly constant (Figure S33). Although the solution pH in a single-cell electrolyzer exceeds 2.6, the oxygen evolution reaction at the anode produces a substantial amount of H^+ (Figure S34). This interfacial microenvironment, enriched in H^+, is conducive to the presence of Co(III) produced by anodic oxidation (Figure S35). The detection of Co(III) also suggests the possibility of oxidizing Co(II) to Co(IV)=O through a two-step single-electron transfer. Besides formation through oxidation, Co(III) can also be derived from the activation of PMS by Co(II). Co(III) may continue to react with the hydrolysis product, the Co(III) dimer ($[Co(III)OH]_2^{4+}$), generated by PMS activation, to convert a binuclear Co species containing a different valence state, which subsequently releases Co(II). This is evidenced by the varying $C_{PMS}/C_{Co(II)}$ ratios, where the disparity between the Co(III)-EDTA signals of the EC/Co/(II)/PMS system and those of the Co/(II)/PMS system diminishes as $C_{PMS}/C_{Co(II)}$ increases (Figure 1c). As shown in Figure 2a, PMS significantly reduces the oxidation potential of Co(II) (discussed in detail in the next section). Consequently, Co(II) is rapidly oxidated to Co(III) at the anode with the aid of high PMS concentrations (pathway V). Subsequently, Co(III) continues to disrupt the $[Co(III)-OH]_2^{4+}$ complex to release Co(II) (pathways III and IV in Figure 1d). At lower PMS concentrations, the reactions in pathways III−V proceed more slowly. However, Co(II)-activated PMS is relatively easy to achieve, allowing Co(III) and $[Co(III)OH]_2^{4+}$ to continually accumulate without reduction. Therefore, the difference in Co(III)-EDTA signals between the EC/Co(II)/PMS and Co(II)/PMS systems is more pronounced at lower $C_{PMS}/C_{Co(II)}$. In fact, with a PMS/Co(II) concentration ratio of 7.5:1, the solution in the EC/Co(II)/PMS system turns yellow after 30 min (which interfered with the determination of Co(III)-EDTA), a state of severe Co hydrolysis.[11] In contrast, as pathway V is enhanced at higher $C_{PMS}/C_{Co(II)}$, it accelerates pathways III and IV, reducing Co(III) accumulation and narrowing the gap between $Abs_{Co(II)/PMS}$ and $Abs_{EC/Co(II)/PMS}$. Notably, the rate of Co(II)/Co(III) cycling driven by the reaction between Co(III) and its hydrolysis products is slower than the conversion of Co(II) to Co(III) by PMS activation. This phenomenon is evident at low $C_{PMS}/C_{Co(II)}$ but is significantly accelerated by increasing the PMS doses, which promote the decomposition of the $[Co(III)OH]_2^{4+}$ complex and the release of Co(II). This conclusion is supported by the Co(III)

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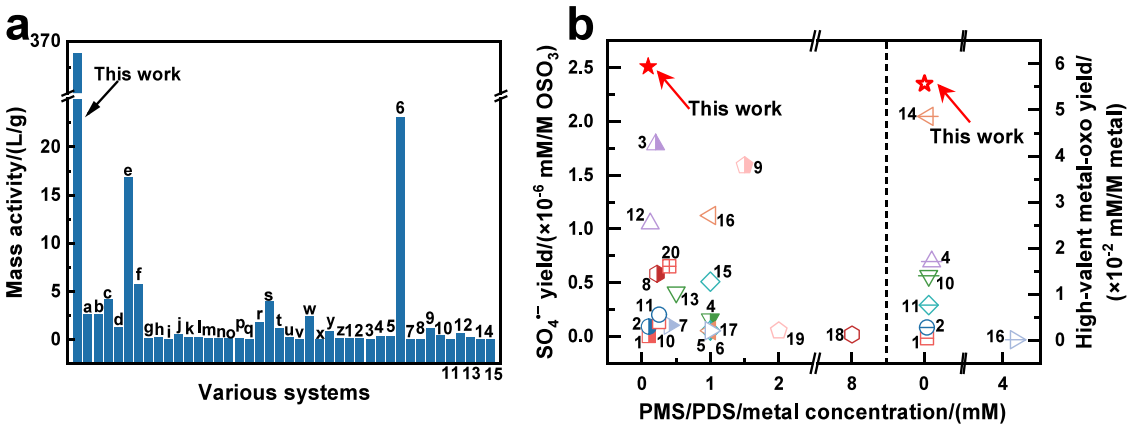


Figure 3. Comparison of Fenton-like systems. (a) Comparison for the mass activity of various catalysts in terms of pollutants degradation per unit mass of catalysts/oxidant. (b) Comparison of $SO_4^{\bullet-}$ and high-valent metal-oxo yields in different Fenton-like processes. Detailed experimental conditions in Tables S5 and S6.

reduction rates observed at different $C_{PMS}/C_{Co(II)}$ ratios. As the $C_{PMS}/C_{Co(II)}$ ratio increased from 7.5:1 to 45:1, the reduction rate significantly increased from 0.0141 to 0.0293 min^{-1} (Figure S31e). Consequently, despite cobalt cycling, the EC/Co(II)/PMS system exhibits a slight decline in CIP removal efficiency over time in stability tests. Nonetheless, in terms of synergistic effects, the synergistic factors of the EC/Co(II)/PMS system in the first 10 cycles are mostly above 2.0 and show an upward trend (Figure 1b). This strong synergistic effect may stem from the cobalt cycle, explaining the stronger Co(III)-EDTA signal in the EC/Co(II)/PMS system compared to the Co(II)/PMS system at the same time point, as Co(II) released from the $[Co(III)OH]_2^{4+}$ complex during electrochemical processes is rapidly reconverted to Co(III) by PMS.

Subsequently, electrochemical tests were conducted to simulate this process, corroborating the aforementioned findings. Analysis of the LSV curves revealed a distinct preference for Co(III) reduction under acidic conditions, whereas reduction was hardly discernible in alkaline solutions (Figure S36). A linear correlation was observed between the Co(III) reduction potential and pH, yielding an E/pH slope of 0.1695 V/pH (Figure S37). According to the Nernst equation, the direct reduction of Co(III) to Co(II) necessitates at least two protons, indicating a proton-coupled electron transfer (PCET) mechanism. Consequently, this process is impeded in cathode regions abundant in OH^-. In CV tests focusing solely on Co(II), a prominent oxidation peak at 0.98 V was identified, attributable to the oxidation of Co(II) to Co(III)/Co(IV) (Figure S38). The introduction of PMS markedly diminished the oxidation potential of Co(II), with this reduction becoming more pronounced as PMS concentration increased. This phenomenon was accompanied by an augmentation of the peak current (Figure 2a and Figure S39). Co(II) electrooxidation proceeded via outer-sphere electron transfer from solvated Co(II) to the anode. The presence of PMS facilitated the oxidation of Co(II) to Co(III) or Co(IV)=O at the anode. Primarily, Co(III) engaged in the oxidation reaction of hydrolysis products, releasing Co(II) as the predominant pathway, while a minor fraction of free Co(III) underwent reduction at the cathode or further oxidation to Co(IV)=O at the anode. Co(IV)=O subsequently reduced to Co(II) concurrent with target pollutant oxidation, thereby establishing a cobalt cycle (Co(II)/Co(III)/Co(IV)) with Co(II) as the active source perpetuating redox reactions. Co(II) played both direct and indirect roles in the generation of $SO_4^{\bullet-}$, •OH, Co(IV)=O, and $^1O_2$. When hydrolyzed Co(III) ($[Co(III)OH]_2^{4+}$) served as the reaction medium, a notable potential increase was evident upon injection of 1 mM free Co(III) (stage II, red line in Figure 2b). This suggests that the electron transfer resulting from redox interactions between $[Co(III)OH]_2^{4+}$ and free Co(III) induced the potential elevation. Following a temporal interval, a renewed surge in potential was observed with the addition of 1 mM PMS (stage III), suggesting that PMS activation by Co(II) released during the preceding stage (stage II) modulated the potential. Conversely, PMS introduction into $[Co(III)OH]_2^{4+}$ solution elicited only gradual, modest potential alterations (purple line in Figure 2b), indicating that $[Co(III)OH]_2^{4+}$ did not activate PMS. In addition, free Co(II) exhibited minimal reactivity toward $[Co(III)OH]_2^{4+}$, resulting in a negligible potential change (blue line in Figure 2b). Notably, visual changes in the reaction solution were observed during the OCPT test (detailed analysis in Figure S40). These observations further substantiate the unique mechanism underlying the Co(II)/Co(III) cycle.

In the reaction involving Co(III) with $[Co(III)OH]_2^{4+}$, both substrates undergo reduction of Co(III) to Co(II). The hydroxyl oxygen, bearing the lone electron pair and responsible for bridging, is postulated to serve as the electron donor, potentially converting the −2 valence oxygen atom to oxygen. Analysis of oxygen concentration evolution in various reaction solutions was conducted (Figure 2c). Neither $[Co(III)OH]_2^{4+}$ nor free Co(II) can produce oxygen independently or through mutual reaction. The emergence of oxygen in solutions containing free Co(III) is attributed to the strong oxidizing ability of Co(III) to oxidize H_2O molecules, generating oxygen. Intriguingly, a significant higher oxygen production was observed in the reaction between $[Co(III)OH]_2^{4+}$ and free Co(III). At the reaction's onset, the oxygen produced (2.615 mg/L) was 18.03 times greater than that from water oxidation by free Co(III) (0.145 mg/L) (Figure S41). This not only demonstrates the rapid nature of the reaction between $[Co(III)OH]_2^{4+}$ and free Co(III) but also corroborates our hypothesis that the electrons on the hydroxyl oxygens in $[Co(III)OH]_2^{4+}$ donate to both the intrinsic Co(III) and exogenous free Co(III), ultimately resulting in oxygen production. *In situ* characterization of the reaction between

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Environ. Sci. Technol. 2025, 59, 19570−19581

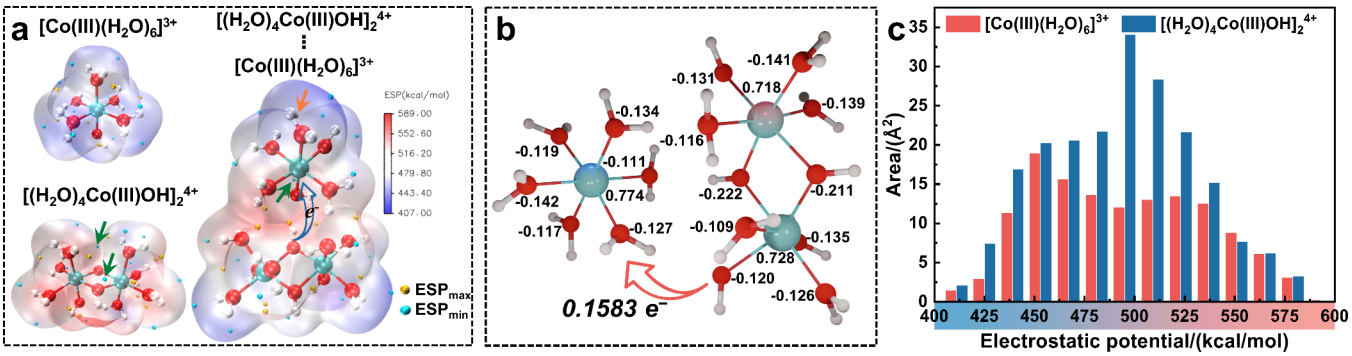


Figure 4. Potential electron transfer pathway. (a) Electrostatic potential (ESP) distribution of freestanding $[Co(III)(H_2O)_6]^{3+}$ and $[(H_2O)_4Co(III)OH]_2^{4+}$ and $[(H_2O)_4Co(III)OH]_2^{4+}\cdots[Co(III)(H_2O)_6]^{3+}$ reactive structure. (b) Hirshfeld charge distribution of the reactive structure and calculated number of transferred electrons. (c) Surface area distribution in different ESP intervals of $[Co(III)(H_2O)_6]^{3+}$ and $[(H_2O)_4Co(III)OH]_2^{4+}$ in the reactive structure.

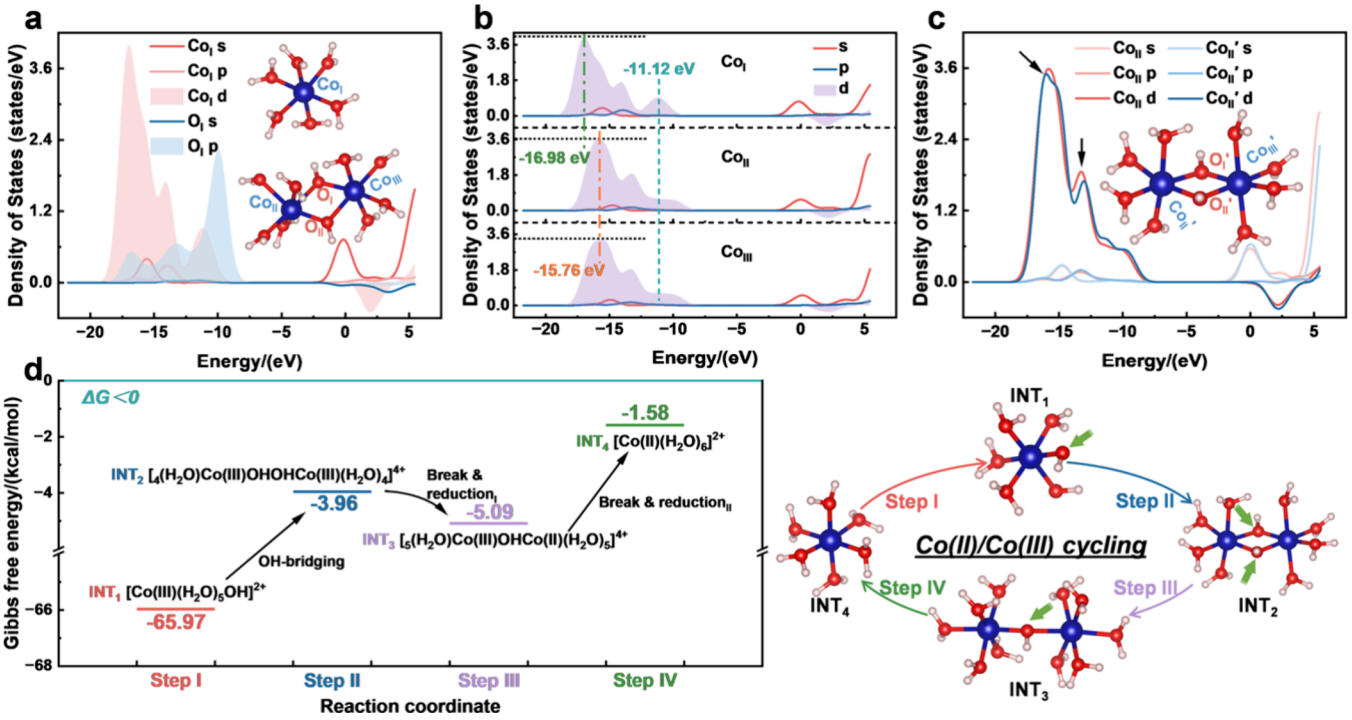


Figure 5. Theoretical calculations and validation. (a) Molecular orbitals and partial density of state (PDOS) of Co_I s, p, and d orbitals and O_I s and p orbitals in the reactive structure. (b) Comparison of the Co s, p, and d orbitals' PDOS at different positions in the reactive structure. (c) Comparison of the Co s, p, and d orbitals' PDOS at the same positions in freestanding $[(H_2O)_4Co(III)OH]_2^{4+}$ and reactive structure. (d) Calculated potential energy diagrams on the Co(II)/Co(III) cyclic processes.

$[Co(III)OH]_2^{4+}$ and free Co(III) is then performed using confocal Raman spectroscopy to confirm the Co(II)/Co(III) cycle (Figure 2d and Figures S42 and S43). Distinct vibrational bands at 980 and 1052 cm^{-1} were observed in the Raman spectrum of Co(II) and free Co(III) solutions, respectively (Figure S43). Upon introducing stabilized $[Co(III)OH]_2^{4+}$ into the quartz flow cell by NaOH injection and subsequently mixing it with free Co(III) at 400 s, a notable decrease in the characteristic Co(III) peaks and a concurrent increase in Co(II) peaks were observed (Figure 2d). This spectral transformation in the Raman characteristics upon mixing provides direct evidence for the reaction dynamics between $[Co(III)OH]_2^{4+}$ and free Co(III), indicating a redox process where Co(III) is reduced to Co(II) and released.

Notably, in terms of catalytic activity in the oxidation reaction, Co(II) in the electrochemical process exhibited a mass activity of 368.87 L/g, surpassing numerous advanced PMS catalysts reported in the literature (Figure 3a). This implies that the EC/Co(II)/PMS process can markedly decrease the requirement for metal salts in heterogeneous catalyst synthesis and reduce oxidant usage in catalytic oxidation. It also demonstrates substantial advantages in minimizing metal leaching. Furthermore, the yields of $SO_4^{\bullet-}$ and Co(IV)=O from HO-OSO_3^- and Co reached 2.51 × 10^{-6} mM/M OSO_3 and 5.57 × 10^{-2} mM/M Co (Figure 3b), respectively, which are relatively high compared to existing research. All of the above may represent a substantial advantage of electrochemical processes and their capacity to drive the Co(II)/Co(III) redox cycle. Therefore, achieving stable and efficient cycling of metal ion catalysts in homogeneous reactions is crucial for sustained pollutant degradation.

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Environ. Sci. Technol. 2025, 59, 19570−19581

Theoretical Calculation and Verification. Density functional theory (DFT) calculations were performed to gain deeper insights into the reaction mechanism between $[Co(III)OH]_2^{4+}$ and Co(III). The electron transfer between these species is pivotal in the cyclic conversion of Co(III) to Co(II), necessitating a thorough examination of their chemical interactions and properties. In our computational model, Co is coordinated with six H_2O molecules to form $[Co(III)(H_2O)_6]^{3+}$, and $[(H_2O)_4Co(III)OH]_2^{4+}$ is generated by substituting one H_2O molecule in $[Co(III)(H_2O)_6]^{3+}$ with a hydroxyl group, resulting in a bridged structure (Figure S44). Our analysis primarily focused on the electrostatic potential (ESP) distribution, Hirshfeld charge, molecular orbitals, and partial density of state (PDOS) between the two species.

The ESP distribution reveals that the maximum (ESP_{max}) and minimum (ESP_{min}) values in the two isolated structures are largely attributed to the positively charged Co(III) ions and H_2O molecules with lone electron pairs, respectively (Figure 4a). In the freestanding $[(H_2O)_4Co(III)OH]_2^{4+}$, the ESP_{min} is localized near the bridged hydroxyl group, which is associated with the lone electron pairs on the hydroxyl oxygen, poised to donate electrons to Co(III). Indeed, upon the approach of $[Co(III)(H_2O)_6]^{3+}$ to $[(H_2O)_4Co(III)OH]_2^{4+}$, the ESP_{min} near the hydroxyl oxygen between the two disappears. Simultaneously, in the $[(H_2O)_4Co(III)OH]_2^{4+}\cdots[Co(III)(H_2O)_6]^{3+}$ reactive structure, the ESP_{max} of $[Co(III)(H_2O)_6]^{3+}$ diminishes significantly, and new ESP_{min} regions emerge around it. As conjectured, electrons from the hydroxyl oxygen in $[(H_2O)_4Co(III)OH]_2^{4+}$ are transferred to $[Co(III)(H_2O)_6]^{3+}$, suggesting a reduction of Co(III) via an apparent outer-sphere electron transfer. The Hirshfeld charge distribution of this reactive structure further confirms the transfer of 0.1583 electron (Figure 4b). A statistical analysis of the surface area distributions across different ESP intervals shows a relatively uniform distribution in the $[(H_2O)_4Co(III)OH]_2^{4+}\cdots[Co(III)(H_2O)_6]^{3+}$ structure (Figure S45). However, the $[(H_2O)_4Co(III)OH]_2^{4+}$ still retains a substantial surface area in the blue electron-rich region (Figure 4c), implying the potential for continuous electron transfer within $[(H_2O)_4Co(III)OH]_2^{4+}$ after interacting with $[Co(III)(H_2O)_6]^{3+}$.

To delve deeper into this conjecture and elucidate the reaction mechanism in terms of molecular orbitals, we performed a partial density of state (PDOS) analysis. The total density of states (TDOS) is predominantly contributed by Co 3d orbitals and O 2p orbitals (Figure S46). The observed different degrees of overlap between Co_I 3d and O_I 2p orbitals suggest orbital hybridization and charge transfer between Co(III) and hydroxyl oxygen during the reaction (Figure 5a). Specifically, hydroxyl oxygen O_I exhibits a higher peak density of states compared to O_{II}, suggesting a greater number of energy levels and bands within this energy range (Figure S47). This may result from electron transfer between $[Co(III)(H_2O)_6]^{3+}$ and $[(H_2O)_4Co(III)OH]_2^{4+}$. In terms of Co, the PDOS of Co at different positions in $[(H_2O)_4Co(III)OH]_2^{4+}\cdots[Co(III)(H_2O)_6]^{3+}$ were compared (Figure 5b). PDOS below the Fermi energy level is found to shift toward higher energies, which causes PDOS to be broadened centered on the Fermi energy level, a phenomenon associated with elevated electronic bandwidth. Moreover, the elevated electronic bandwidth often results from enhanced hybridization interactions between orbitals.[45] Notably, Co_I displays a higher PDOS at −16.98 eV and a novel peak at −11.12 eV compared to Co_{II} and Co_{III} (Figure 5b). Combined molecular orbital analysis indicates complex orbital hybridization and electron transfer between Co_I and O_I. As Co_{II} and Co_{III} are bridged by O_I and O_{II}, electron transfer between Co_I and O_I could disrupt the electron distribution around Co_{II} and Co_{III}, affecting their interactions. Further exploration of the PDOS and d-band centers (ε_d) of Co at the same positions in isolated $[(H_2O)_4Co(III)OH]_2^{4+}$ and $[(H_2O)_4Co(III)OH]_2^{4+}\cdots[Co(III)(H_2O)_6]^{3+}$ structures reveals that, similar to O_I and O_{II}, the Co_{II} 2p orbital in the reactive structure exhibits a higher PDOS than the Co_{II}' 2p orbital (Figure 5c and Figure S48). This suggests that the reaction between Co_I and O_I indeed induces a redistribution of electrons around Co_{II}. The electron-rich hydroxyl oxygen may further donate electrons to Co_{II}, resulting in an increased PDOS for the Co_{II} 2p orbital. Additionally, the ε_d of Co_{II} and Co_{III} in the $[(H_2O)_4Co(III)OH]_2^{4+}\cdots[Co(III)(H_2O)_6]^{3+}$ structure shift toward higher energies, with closer proximity to the Fermi level indicating the higher position of the antibonding state and the stronger interaction with hydroxyl oxygens, which may be related to the redistribution of electrons at the heterogeneous interface.[46,47] This represents an inner-sphere electron transfer within the structure. In conclusion, $[Co(III)(H_2O)_6]^{3+}$ and $[(H_2O)_4Co(III)OH]_2^{4+}$ facilitate the reduction of Co(III) through an outer-sphere electron transfer. This process disrupts the electron distribution among O_I, Co_{II}, and Co_{III} and reconfigures them, with the inner-sphere electron transfer potentially leading to the reduction and release of Co from the hydrolyzed complex.

Ultimately, the plausibility of the Co cycle chain reaction process was substantiated through Gibbs free energy calculations (Figure 5d). Initially, upon PMS activation, $[Co(II)(H_2O)_6]^{2+}$ undergoes conversion to $[Co(III)(H_2O)_6]^{3+}$, which readily undergoes hydrolysis and subsequently binds with OH^- to produce $[Co(III)(H_2O)_5OH]^{2+}$ (step I, eq 1). This intermediate then transforms into the dimeric species $[_4(H_2O)Co(III)OHOHCo(III)(H_2O)_4]^{4+}$ ($[(H_2O)_4Co(III)OH]_2^{4+}$) via an OH^--bridged linkage (step II, eq 2). Subsequently, $[Co(II)(H_2O)_6]^{2+}$ is oxidized to $[Co(III)(H_2O)_6]^{3+}$ at the electrochemical anode. $[Co(III)(H_2O)_6]^{3+}$ subsequently engages in electron transfer with the bridging hydroxyl oxygen within $[(H_2O)_4Co(III)OH]_2^{4+}$, resulting in the reduction of Co(III) and the liberation of oxygen (step III, eq 3). An analogous redox process occurs with the alternative bridging hydroxyl oxygen (step IV, eq 4), and these steps III and IV collectively facilitate the reduction and release of Co(III) from the dimer. Notably, the ΔG values of four steps involved in the Co cycle are negative, although the degrees of exothermicity vary. This indicates that these reactions are spontaneous and thermodynamically favorable. Furthermore, the presence of an electric field as an external energy source undoubtedly facilitates the occurrence of these reactions. Drawing upon both experimental data and DFT calculations, we elucidate the Co cycling mechanism within the electrochemical system (EC/Co(II)/PMS). Here, Co(III) generated through electrochemical anodic oxidation and the hydrolyzed dimer $[Co(III)OH]_2^{4+}$ contribute to the reduction and release of Co(III) via outer- and inner-sphere electron transfer processes, accompanied by the emission of oxygen.

$$[Co(III)(H_2O)_6]^{3+} + OH^- \rightarrow [Co(III)(H_2O)_5OH]^{2+} + H_2O \quad (1)$$

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Environ. Sci. Technol. 2025, 59, 19570−19581

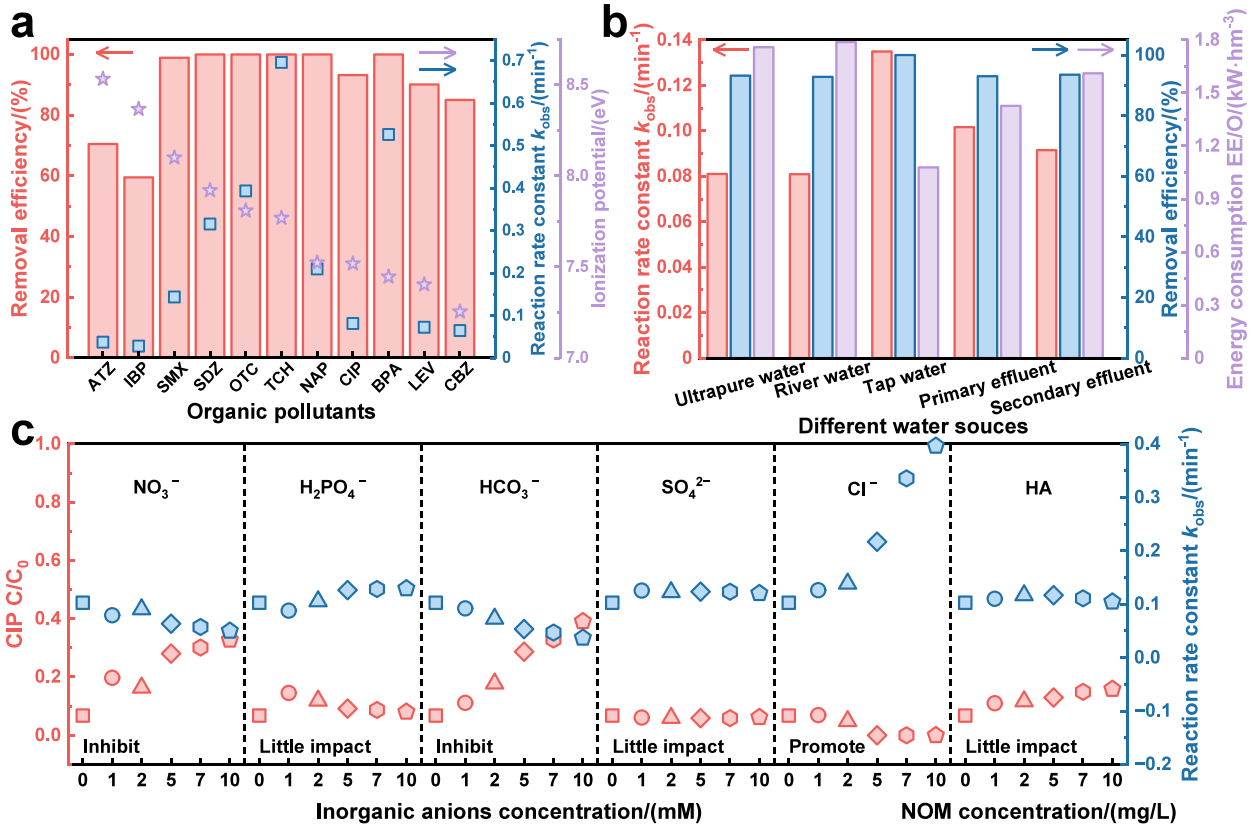


Figure 6. Preliminary exploration for the practicality of the EC/Co(II)/PMS process. (a) Correspondence between removal efficiency, ionization potential, and reaction rate constants k_{obs} of different organic pollutants. Effect of different (b) water bodies and (c) background ions on CIP degradation by the EC/Co(II)/PMS process. Reaction conditions: [pollutant] = 15 μM, [PMS] = 90 μM, Co(II) = 6 μM, Na_2SO_4 = 50 mM, current density = 10 mA/cm^2, T = 293 K, initial solution pH = 6.00 ± 0.05.

$$[Co(III)(H_2O)_5OH]^{2+} + [Co(III)(H_2O)_5OH]^{2+} \rightarrow [_4(H_2O)Co(III)OHOHCo(III)(H_2O)_4]^{4+} + 2H_2O \quad (2)$$

$$[_4(H_2O)Co(III)OHOHCo(III)(H_2O)_4]^{4+} + [Co(III)(H_2O)_6]^{3+} + 2H_2O \rightarrow [Co(II)(H_2O)_6]^{2+} + [_5(H_2O)Co(III)OHCo(II)(H_2O)_5]^{4+} + 1/2O_2 + H^+ \quad (3)$$

$$[_5(H_2O)Co(III)OHCo(II)(H_2O)_5]^{4+} + [Co(III)(H_2O)_6]^{3+} + 2H_2O \rightarrow 2[Co(II)(H_2O)_6]^{2+} + [Co(II)(H_2O)_6]^{2+} + 1/2O_2 + H^+ \quad (4)$$

Potential for Practical Water Purification Applications. To thoroughly assess the practical applicability of the EC/Co(II)/PMS process in real-world purified water scenarios, we conducted a comprehensive evaluation encompassing ecological safety, broad-spectrum efficacy, and resistance to interferences, alongside testing its decontamination capabilities in natural aquatic environments. Prior to widespread implementation, verifying the ecological safety of water purification technologies is imperative to prevent secondary pollution. The degradation pathway and mechanism of CIP in the EC/Co(II)/PMS system were elucidated through UPLC-QTOF-MS/MS analysis, augmented by Fukui index calculations. Degradation predominantly targeted the quinolone and piperazine rings of CIP, involving hydroxylation, denitration, elimination of aldehyde and ethylamine groups, amino hydrolysis, and oxidative ring opening. These transformations are attributed to the concerted action of ROS (detailed analysis in Figures S49−S57). Utilizing the toxicity estimation software tool (TEST), which is grounded in quantitative structure−activity relationship (QSAR), we predicted the acute toxicity (ACT), developmental toxicity, bioaccumulation potential, and mutagenicity of CIP and its degradation intermediates (Figure S58). The findings affirmed the safety and environmental benignity of the process. This conclusion was further corroborated by evaluating the biotoxicity of the treated solutions. Specifically, we monitored the fluorescence intensity of *Vibrio fischeri* and the growth kinetics of *Escherichia coli* (*E. coli*) in time-series samples. Given that antibiotics exhibit potent antibacterial activity against Gram-negative bacteria, the notable increase in *Vibrio fischeri* fluorescence over time indicated a reduction in ACT (Figure S59). Moreover, the earlier onset of stable *E. coli* growth in medium containing degradation intermediates provided further evidence for the safety of the EC/Co(II)/PMS system (Figures S60 and S61).

To investigate the degradation selectivity of the EC/Co(II)/PMS process, organic compounds with diverse ionization potential (IP) were selected for analysis (Figure S62). Our findings suggest the existence of an approximate threshold at 8.3665 eV, corresponding to the IP of ibuprofen (IBP) (Figure

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Environ. Sci. Technol. 2025, 59, 19570−19581

6a). Below this threshold, the process favors the capture of electrons from organic matter by ROS, demonstrating heightened degradation efficacy toward compounds such as sulfamethoxazole (SMX), sulfadiazine (SDZ), oxytetracycline (OTC), tetracycline hydrochloride (TCH), naproxen (NAP), ciprofloxacin (CIP), bisphenol A (BPA), levofloxacin (LEV), and carbamazepine (CBZ). Conversely, organic compounds with high IPs, exemplified by atrazine (ATZ) and IBP, are difficult to lose electrons and consequently resistant to efficient oxidation. Nonetheless, the EC/Co(II)/PMS process exhibits remarkable degradation proficiency across a broad array of organic pollutants.

The oxidation process, predominantly the nonradical pathway, demonstrates robust resilience against interference from environmental constituents (Figure 6c). Specifically, $H_2PO_4^-$, SO_4^{2-}, and humic acid (HA) exert minimal influence on CIP degradation, with $H_2PO_4^-$ further contributing to optimal system performance by modulating pH to its favorable range. SO_4^{2-}, as an inert ion, virtually does not participate in electrochemical reactions. In contrast, the inhibitory effects of HCO_3^- on CIP degradation stem from two aspects: one is the hydrolysis of Co(II) caused by interfering with the reaction pH, and the other is its competitive reaction with CIP degradation. Notably, Cl^- undergoes electrochemical and ROS-mediated transformation into active chlorine species, including $Cl^•$, HOCl, $Cl_2^{•-}$, $HOCl^{•-}$, and $ClO^•$, which accelerates CIP removal.[48] Conversely, NO_3^- is reduced to ammonia at the cathode,[49] competing with Co(III) and PMS for reduction and electrical activation, respectively. Even under high-salt conditions (100 mM level), this electrochemical system exhibits remarkably stable degradation performance (Figure S65). A comprehensive examination of these interactions is provided in Figures S63−S65. Furthermore, the EC/Co(II)/PMS process was evaluated in natural waters and wastewater matrices (Figure 6b and Figure S66; detailed water quality parameters in Table S7). The results were encouraging, with CIP achieving near-complete removal within 30 min. Intriguingly, the varied water sources did not increase energy consumption, but rather, they contributed to a modest reduction. These outcomes underscore the exceptional practical applicability and robustness of the EC/Co(II)/PMS system in real-world scenarios. The EC/Co(II)/PMS system demonstrates continuous capability for treating antibiotic wastewater. Under continuous flow conditions, the CIP removal efficiency gradually increases with extended hydraulic retention time (HRT), and the effluent quality stabilizes without significant fluctuations (Figure S67 and S68). Compared with other Fenton-like processes, the EC/Co(II)/PMS system exhibits significant advantages in high-TOC-load removal, with a normalized indicator of 2.478 L/(mmol h) (Table S8 and Figures S69 and S70). This indicates that the electric field-induced accelerated Co(II)/Co(III) redox cycling can sustainably activate PMS, thereby enhancing the mineralization efficiency of organic pollutants. Furthermore, this system also demonstrated excellent performance in the removal of COD from real antibiotic-containing wastewater, showcasing its robust oxidation and mineralization capabilities. Likewise, it proved highly effective in wastewater decolorization (Figure S71). These collective results verified the potential for practical application of the EC/Co(II)/PMS system. Life cycle assessment (LCA) was conducted on four systems to comprehensively understand the environmental impacts of the product throughout its entire life cycle. The analysis results indicate that the EC/Co(II)/PMS system stood out as the most environmentally sustainable system (Tables S9−S13 and Figure S72; detailed analysis in Figure S73).

■ ENVIRONMENTAL IMPLICATION

In the present study, we successfully achieved efficient and sustained activation of PMS by trace Co(II) using electrochemistry as a cocatalyst. Our results underscore the pivotal role of Co(II) as an active source for catalytic reaction. The electrochemical regeneration of Co(II) sustained the catalytic oxidation reaction, a feat unattainable with Co(II) alone. We conducted an in-depth analysis of the unique Co cycling mechanism during the electrochemical processes. Co(III), generated through anodic oxidation, emerged as a key species in the disruption of the hydrolyzed dimer complexes. This allowed Co(III) to undergo reduction via both outer- and inner-sphere electron transfer mechanisms. Specifically, the lone electron pairs of the hydroxyl oxygen, which serve as bridges in the $[Co(III)OH]_2^{4+}$ dimer, are transferred to Co(III) through outer-sphere electron transfer. This process disrupts the electron distribution between O and Co, facilitating the subsequent reduction of Co(III) within the dimer via inner-sphere electron transfer. Furthermore, our system demonstrated remarkable performance across a variety of environmental conditions, effectively degrading a wide spectrum of pollutants while exhibiting good ecological safety and resistance to external disturbances. This work not only provides a solution to the rate-limiting step of $M^{(n+1)+}$ reduction in advanced oxidation processes catalyzed by homogeneous transition metal (M^{n+}) but also offers a novel perspective on the sustainability of catalytic oxidation. By emphasizing the particular mechanism of Co recycling during electrochemical processes, we have addressed the challenge posed by the easy hydrolysis and reduction resistance of Co(III) formed after Co(II) conversion by PMS, thereby ensuring the continuous and efficient operation of the catalytic system.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c03860.

Detailed description of chemicals; experimental procedures; DFT calculation; analytical methods; LCA analysis; electrical energy consumption; performance comparison of different systems; water quality parameters; CIP degradation pathways; biological toxicity assessment; electrochemical system performance test; identification, steady-state concentration, and oxidation contribution calculation of ROS; proposal and validation of the cyclic mechanism; potential practical applicability (PDF)

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Notes

The authors declare no competing financial interest.

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